

# LAN Simulation Project Implementation Guide

Version 2.0: Enhanced Network with Services

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**Version Note:** This guide covers **Version 2.0** implementation with wireless access, DHCP server, mail server, and enhanced DNS configuration.

## Phase 1: Topology Physical Construction v2

### 1. Connections

#### 1.1 SW1 connections with Router, DNS, Mail Server, Access Points

R1 G0/0 then with SW1 G0/1

SW2 G0/1 then with SW1 G0/2

DNS1 Fa0 then with SW1 Fa0/24 (**Corrected**)

MAIL1 Fa0 then with SW1 Fa0/10 (**v2 Addition**)

AP-CS Fa0 then with SW1 Fa0/8 (**v2 Addition - CS Dept WiFi**)

AP-ENG Fa0 then with SW1 Fa0/9 (**v2 Addition - ENG Dept WiFi**)

#### 1.2 SW1 connections with PC1

CS1 fa0 then with SW1 fa0/3

ENG1 fa0 then with SW1 fa0/4

BUS1 fa0 then with SW1 fa0/5

BIO1 fa0 then with SW1 fa0/6

#### 1.3 SW2 connections PCs2

CS2 fa0 then with SW2 fa0/2

ENG2 fa0 then with SW2 fa0/3

BUS2 fa0 then with SW2 fa0/4

BIO2 fa0 then with SW2 fa0/5

## 1.4 Wireless Devices (v2 Addition)

Laptop-CS connected wirelessly to AP-CS (SSID: CS\_Dept\_WiFi)

Laptop-ENG connected wirelessly to AP-ENG (SSID: ENG\_Dept\_WiFi)

# Phase 2: Configurations SW, Access ports, trunk ports v2

## 2 Configuration SW1; Access ports assign, Trunk Ports

### 2.1 Configure SW1 First

```
Switch> enable
Switch# configure terminal

Switch(config)# vlan 10
Switch(config-vlan)# name CS_Dept
Switch(config-vlan)# exit

Switch(config)# vlan 20
Switch(config-vlan)# name ENG_Dept
Switch(config-vlan)# exit

Switch(config)# vlan 30
Switch(config-vlan)# name BUS_Dept
Switch(config-vlan)# exit

Switch(config)# vlan 40
Switch(config-vlan)# name BIO_Dept
Switch(config-vlan)# exit

Switch(config)# vlan 99
Switch(config-vlan)# name SERVERS
Switch(config-vlan)# exit
```

### 2.2 Access Port Assignments on SW1 (v2 Enhanced)

```
! CS1 to VLAN 10 (Fa0/3)
Switch(config)# interface FastEthernet0/3
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# exit

! ENG1 to VLAN 20 (Fa0/4)
Switch(config)# interface FastEthernet0/4
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
```

```
Switch(config-if)# exit

! BUS1 to VLAN 30 (Fa0/5)
Switch(config)# interface FastEthernet0/5
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 30
Switch(config-if)# exit

! BIO1 to VLAN 40 (Fa0/6)
Switch(config)# interface FastEthernet0/6
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 40
Switch(config-if)# exit

! DNS1 to VLAN 99 (Fa0/24)
Switch(config)# interface FastEthernet0/24
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 99
Switch(config-if)# exit

! MAIL1 to VLAN 99 (Fa0/10) - v2 Addition
Switch(config)# interface FastEthernet0/10
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 99
Switch(config-if)# exit

! AP-CS to VLAN 10 (Fa0/8) - v2 Addition
Switch(config)# interface FastEthernet0/8
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# exit

! AP-ENG to VLAN 20 (Fa0/9) - v2 Addition
Switch(config)# interface FastEthernet0/9
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# exit
```

## 2.3 Trunk ports on SW1

```
! Gi0/1 to Router as TRUNK
Switch(config)# interface GigabitEthernet0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit

! Gi0/2 to SW2 as TRUNK
Switch(config)# interface GigabitEthernet0/2
```

```
Switch(config-if)# switchport mode trunk
Switch(config-if)# end
```

### 3 Configuration SW2; Access ports assign, Trunk Ports

#### 3.1 configure SW2

```
! Access SW2 CLI tab
Switch> enable
Switch# configure terminal

! Create VLANs (same as SW1)
Switch(config)# vlan 10
Switch(config-vlan)# name CS_Dept
Switch(config-vlan)# exit

Switch(config)# vlan 20
Switch(config-vlan)# name ENG_Dept
Switch(config-vlan)# exit

Switch(config)# vlan 30
Switch(config-vlan)# name BUS_Dept
Switch(config-vlan)# exit

Switch(config)# vlan 40
Switch(config-vlan)# name BIO_Dept
Switch(config-vlan)# exit
```

#### 3.2 Access Port Assignments on SW2

```
! CS2 (Fa0/2)      VLAN 10
Switch(config)# interface FastEthernet0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# exit

! ENG2 (Fa0/3)     VLAN 20
Switch(config)# interface FastEthernet0/3
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# exit

! BUS2 (Fa0/4)     VLAN 30
Switch(config)# interface FastEthernet0/4
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 30
```

```
Switch(config-if)# exit

! BIO2 (Fa0/5)      VLAN 40
Switch(config)# interface FastEthernet0/5
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 40
Switch(config-if)# exit
```

### 3.3 Trunk ports on SW2

```
! Configure Gi0/1 as TRUNK (to SW1)
Switch(config)# interface GigabitEthernet0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# end

! Verify configuration
Switch# show vlan brief
```

## Phase 3: Router Configuration (Inter-VLAN Routing v2)

```
! Enter privileged mode
Router> enable
Router# configure terminal

! Configure main interface (no IP address here)
Router(config)# interface GigabitEthernet0/0
Router(config-if)# no shutdown
Router(config-if)# no ip address
Router(config-if)# exit

! Create subinterface for VLAN 10 (CS_Dept)
Router(config)# interface GigabitEthernet0/0.10
Router(config-subif)# description Gateway for CS_Dept VLAN
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
! v2 Addition: DHCP Relay for VLAN 10
Router(config-subif)# ip helper-address 192.168.99.10
Router(config-subif)# exit

! Create subinterface for VLAN 20 (ENG_Dept)
Router(config)# interface GigabitEthernet0/0.20
Router(config-subif)# description Gateway for ENG_Dept VLAN
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
```

```
! v2 Addition: DHCP Relay for VLAN 20
Router(config-subif)# ip helper-address 192.168.99.10
Router(config-subif)# exit

! Create subinterface for VLAN 30 (BUS_Dept)
Router(config)# interface GigabitEthernet0/0.30
Router(config-subif)# description Gateway for BUS_Dept VLAN
Router(config-subif)# encapsulation dot1Q 30
Router(config-subif)# ip address 192.168.30.1 255.255.255.0
! v2 Addition: DHCP Relay for VLAN 30
Router(config-subif)# ip helper-address 192.168.99.10
Router(config-subif)# exit

! Create subinterface for VLAN 40 (BIO_Dept)
Router(config)# interface GigabitEthernet0/0.40
Router(config-subif)# description Gateway for BIO_Dept VLAN
Router(config-subif)# encapsulation dot1Q 40
Router(config-subif)# ip address 192.168.40.1 255.255.255.0
! v2 Addition: DHCP Relay for VLAN 40
Router(config-subif)# ip helper-address 192.168.99.10
Router(config-subif)# exit

! Create subinterface for VLAN 99 (SERVERS)
Router(config)# interface GigabitEthernet0/0.99
Router(config-subif)# description Gateway for SERVERS VLAN
Router(config-subif)# encapsulation dot1Q 99
Router(config-subif)# ip address 192.168.99.1 255.255.255.0
! v2 Addition: DHCP Relay for VLAN 99
Router(config-subif)# ip helper-address 192.168.99.10
Router(config-subif)# exit

! Return to privileged mode
Router(config-subif)# end
```

## Phase 4: Network Services Configuration v2

### 4.1 Configure DNS Server

IP Address: 192.168.99.10  
Subnet Mask: 255.255.255.0  
Default Gateway: 192.168.99.1  
DNS Server: 192.168.99.10 (itself)

### 4.2 Configure DNS Service (v2 Enhanced)

go to Services tab → DNS

Click DNS Service: ON

**Add A Records:**

|                |                        |
|----------------|------------------------|
| Name: cs1.lan  | Address: 192.168.10.10 |
| Name: cs2.lan  | Address: 192.168.10.20 |
| Name: eng1.lan | Address: 192.168.20.10 |
| Name: eng2.lan | Address: 192.168.20.20 |
| Name: bus1.lan | Address: 192.168.30.10 |
| Name: bus2.lan | Address: 192.168.30.20 |
| Name: bio1.lan | Address: 192.168.40.10 |
| Name: bio2.lan | Address: 192.168.40.20 |
| Name: dns1.lan | Address: 192.168.99.10 |

**v2 Additions:**

|                           |                         |
|---------------------------|-------------------------|
| Name: mail.university.lan | Address: 192.168.99.20  |
| Name: university.lan      | Address: 192.168.99.10  |
| Name: laptop-cs.lan       | Address: 192.168.10.101 |
| Name: laptop-eng.lan      | Address: 192.168.20.101 |

### 4.3 Configure DHCP Server on DNS1 (v2 New Feature)

go to Services tab → DHCP

Click DHCP Service: ON

**Create DHCP Pools:**

Pool Name: VLAN10\_CS  
Default Gateway: 192.168.10.1  
DNS Server: 192.168.99.10  
Start IP: 192.168.10.50  
Subnet Mask: 255.255.255.0  
Max Users: 25

Pool Name: VLAN20\_ENG  
Default Gateway: 192.168.20.1  
DNS Server: 192.168.99.10  
Start IP: 192.168.20.50  
Subnet Mask: 255.255.255.0  
Max Users: 25

Pool Name: VLAN30\_BUS  
Default Gateway: 192.168.30.1  
DNS Server: 192.168.99.10  
Start IP: 192.168.30.50  
Subnet Mask: 255.255.255.0  
Max Users: 25

Pool Name: VLAN40\_BIO

Default Gateway: 192.168.40.1  
DNS Server: 192.168.99.10  
Start IP: 192.168.40.50  
Subnet Mask: 255.255.255.0  
Max Users: 25

#### 4.4 Configure Mail Server MAIL1 (v2 New Feature)

IP Address: 192.168.99.20  
Subnet Mask: 255.255.255.0  
Default Gateway: 192.168.99.1  
DNS Server: 192.168.99.10  
**Mail Service Configuration:** go to Services tab → MAIL  
Click MAIL Service: ON  
Domain Name: university.lan  
**Add User Accounts:**

User: student1 Password: student123  
User: professor1 Password: prof123  
User: admin1 Password: admin123

#### 4.5 Configure Wireless Access Points (v2 New Feature)

**For AP-CS:** Config tab → INTERFACE: Wireless0 SSID: CS\_Dept\_WiFi Authentication: WPA2-PSK PSK Pass Phrase: CSdept2026

**For AP-ENG:** Config tab → INTERFACE: Wireless0 SSID: ENG\_Dept\_WiFi Authentication: WPA2-PSK PSK Pass Phrase: ENGdept2026

#### 4.6 Configure Wireless Laptops (v2 New Feature)

**For Laptop-CS:** 1. Physical tab: Install PT-LAPTOP-NM-1W wireless card 2. Desktop tab → PC Wireless → Connect to CS\_Dept\_WiFi 3. Password: CSdept2026 4. IP Configuration → DHCP

**For Laptop-ENG:** 1. Physical tab: Install PT-LAPTOP-NM-1W wireless card 2. Desktop tab → PC Wireless → Connect to ENG\_Dept\_WiFi 3. Password: ENGdept2026 4. IP Configuration → DHCP

#### 4.7 Configure DHCP Clients (v2 Enhancement)

**Change these devices to DHCP:** - CS2, ENG2, BUS2, BIO2 - Laptop-CS, Laptop-ENG

**Method:** Desktop tab → IP Configuration → Select DHCP

**Expected IP assignments:** - CS2: 192.168.10.50 - ENG2: 192.168.20.50 - Laptop-CS: 192.168.10.51 - Laptop-ENG: 192.168.20.51